AMT Worksheet 12
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The purpose of this assignment is to illustrate some of the different types of collections of objects seen in mathematics, specifically fields, rings, groups, and vector spaces.

Determine whether the following sets form a group under the indicated operation. If it does not, explain why not and give an explicit example illustrating this failure. If it does, explain why and tell whether it is an abelian group or not.

1. The set of symmetric 2×2 matrices, with integer entries, under addition.

2. The set of symmetric 2×2 matrices, with integer modulo 5 entries, under addition.

3. The set of symmetric 2×2 matrices, with integer entries, under multiplication.

4. The set of symmetric matrices $\begin{pmatrix} a & b \\ b & d \end{pmatrix}$, with $a,b,d \in \mathbb{Z}_5$ (the set of integers modulo 5) and $ad-b^2 \neq 0$, under multiplication.

5. The set of matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, with $a,b,c,d \in \mathbb{Z}_5$ (the set of integers modulo 5) and $ad-bc \neq 0$, under multiplication.

6. The set of matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, with $a,b,c,d \in \mathbb{Z}_6$ (the set of integers modulo 6) and $ad-bc \neq 0$, under multiplication.

7. The set of numbers $a+b\sqrt{2}$, with $a,b \in \mathbb{Z}$, under addition.

8. The set of numbers $a+b\sqrt{2}$, with $a,b \in \mathbb{Z}$ and $a^2+b^2 \neq 0$, under multiplication.

9. The set of numbers $a+b\sqrt{2}$, with $a,b \in \mathbb{Z}_3$, under addition.

10. The set of numbers $a+b\sqrt{2}$, with $a,b\in\mathbb{Z}_3$ and $a^2+b^2\neq 0$, under multiplication.

11. The set of numbers $a+b\sqrt{2}$, with $a,b\in\mathbb{Z}_6$, under addition.

12. The set of numbers $a+b\sqrt{2}$, with $a,b\in\mathbb{Z}_6$ and $a^2+b^2\neq 0$, under multiplication.

13. The set of numbers $a+b\ln 2$, with $a,b\in\mathbb{Q}$, under addition.

14. The set of numbers $a+b\ln 2$, with $a,b\in\mathbb{Q}$ and $a^2+b^2\neq 0$, under multiplication.

15. The set of $n\times n$ matrices M with real entries satisfying $M^TM=I$, under multiplication.

16. The set of $n\times n$ matrices M with real entries satisfying $M^TM=I$ with $\det(M)=+1$, under multiplication.

17. The set of $n\times n$ matrices M with real entries satisfying $M^TM=I$ with $\det(M)=-1$, under multiplication.

18. The set of integer powers of 2 under multiplication.

Determine whether the following sets form a ring under ordinary addition and multiplication. If it does not, explain why not and give an explicit example illustrating this failure. If it does, explain why and tell whether it is an abelian ring or not.

19. The set of 2×2 matrices with integer entries.

20. The set of matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, with $a,b,c,d\in\mathbb{Z}_5$.

21. The set of matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, with $a,b,c,d\in\mathbb{Z}_6$.

22. The set of numbers $a+b\sqrt{2}$, with $a,b\in\mathbb{Q}$.

23. The set of numbers $a+b\ln 2$, with $a,b\in\mathbb{Q}$.

24. The set of numbers $a+b\sqrt{2}$, with $a,b\in\mathbb{Z}_3$.

25. The set of numbers $a+b\sqrt{2}$, with $a,b\in\mathbb{Z}_5$.

26. The set of numbers $a+b\sqrt{2}$, with $a,b\in\mathbb{Z}_6$.

27. The set of $n\times n$ matrices M with real entries satisfying $M^TM=I$.

Determine whether the following sets form a field under ordinary addition and multiplication. If it does not, explain why not and give an explicit example illustrating this failure. If it does, explain why.

28. The set of 2×2 matrices with integer entries.

29. The set of matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, with $a,b,c,d\in\mathbb{Z}_5$ and $ad-bc\neq 0$.

30. The set of matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, with $a,b,c,d\in\mathbb{Z}_6$ and $ad-bc\neq 0$.

31. The set of numbers $a+b\sqrt{2}$, with $a,b\in\mathbb{Q}$.

32. The set of numbers $a+b\sqrt{2}$, with $a,b\in\mathbb{Z}_3$.

33. The set of numbers $a+b\sqrt{2}$, with $a,b\in\mathbb{Z}_5$.

34. The set of numbers $a+b\sqrt{2}$, with $a,b\in\mathbb{Z}_6$.

35. The set of all invertible 2×2 matrices with entries in $\mathbb{Z}_3$ forms a finite group under multiplication. How many elements does it contain? List 5 of these that are their own inverse and 3 pairs of matrix and inverse. Is this group abelian? Explain.

36. Consider the set of numbers $a+b\sqrt{2}+c\sqrt{3}$, with $a,b,c\in\mathbb{Q}$.
    (a) Explain why this set does not form a field.

    (b) Based on your answer to part (a) suggest an extension, that is, a larger set of numbers containing this set, that is a field. Make your extension as minimal as possible, so that it just solves the problem you laid out in part (a) and doesn't add any unnecessary elements.

    (c) Would you be able to extend the set $a+b\sqrt{5}+c\sqrt{7}+d\sqrt{17}$ in a similar way? Explain why or why not.

    (d) Would you be able to extend the set $a+b\sqrt{2}+c\sqrt[3]{2}$ in a similar way? Explain why or why not.

    (e) Would you be able to extend the set $a+b\ln 2$ in a similar way? Explain why or why not.

    (f) Would you be able to extend the set $a+b\,\pi$ in a similar way? Explain why or why not.

    (g) The *degree* of a field can, for our purposes, be thought of as the number of letters required in an expression like $a+b\sqrt{2}+c\sqrt{3}$ to write a general element of the field. Give the degree of the field you discussed in each of parts (b) – (f).